



The use of statins can reduce the risk of osteoarthritis and osteoporosis

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Abstract

Background Observational studies suggest that statins may be protective against osteoporosis and osteoarthritis. However, the association between statins and the risk of osteoporosis and osteoarthritis has not been fully clarified.

Objective The association between statins and the risk of osteoporosis and osteoarthritis was explored by Mendelian randomization analyses using pooled data from a large population-based genome-wide association study (GWAS).

Methods Data on statin-associated single-nucleotide polymorphisms were obtained from version 10 of the FinnGen database, while information regarding osteoporosis and osteoarthritis was sourced from IEU Open GWAS and UK Biobank, database. The primary analytical method employed was inverse variance weighting to assess causal effects. Additional methods including weighted median, simple median, weighted mode, and MR-Egger regression were utilized for further clarification. Odds ratios along with 95% confidence intervals were calculated to evaluate the causal relationship between statins and the risks associated with osteoarthritis and osteoporosis. Subsequently, sensitivity analyses were conducted to verify result reliability by employing Cochran's Q test for heterogeneity assessment, MR-Egger intercept for detecting horizontal pleiotropy presence, as well as leave-one-out analysis to identify potentially influential single nucleotide polymorphisms.

Results (1) The results of the inverse variance weighting method showed a negative causal relationship between genetically predicted statins and osteoporosis (OR = 0.998, 95% CI 0.996–0.999, $P = 0.01$) risk; and a negative causal relationship between statins and osteoarthritis (OR = 0.974, 95% CI 0.951–0.998, $P = 0.037$) risk; (2) MR-Egger-intercept analysis did not detect potential horizontal pleiotropy (osteoporosis: $P = 0.848$; osteoarthritis: $P = 0.483$); and (3) the findings provide favorable evidence that statins reduce the risk of osteoporosis and osteoarthritis as described in the observational study, but further research is needed on the specific mechanisms of statins for osteoporosis and osteoarthritis.

Conclusion Statins reduce the risk of osteoporosis and osteoarthritis, which is consistent with the results of observational studies. Further studies are essential to validate these results and explore the underlying mechanisms in detail.

Keywords Statin · Osteoporosis · Osteoarthritis · Mendelian randomization · Causality

Introduction

Osteoporosis (OP) is a common systemic skeletal disease in the population, associated with aging, and is characterized by a decrease in bone mineral density (BMD), leading to increased bone fragility, characterized by increased fracture risk in older adults (Sözen et al. 2017). It causes significant emotional, physical, and financial stress to patients and often leads to disability and reduced quality of life (Lane 2006)

Osteoarthritis (OA) is the most common joint disease, where degenerative changes in articular cartilage lead to aseptic inflammation and involvement of the entire joint tissue (Glyn-Jones et al. 2015), and is a major cause of reduced quality of life in older adults. As the population ages, it is

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expected that an increasing number of middle-aged and older adults will be affected by large joint arthritis (Jiang 2022). OA primarily causes morning stiffness, joint pain and pressure, and swelling or deformity of the joints, which reduces the patient's quality of life (Mandl 2019). Risk factors that have been identified as contributing to OA include age, gender, mechanical factors, genetic predisposition and obesity (Lane 2006). However, the exact etiology and pathological changes of OA remain unclear, leading to a lack of curable treatments.

Hydroxy-3-methylglutaryl coenzyme A reductase inhibitors (statins) have become the mainstay of medications for the prevention and treatment of cardiovascular disease (CVD), and statins appear to be potentially promising agents for the treatment of osteoporosis (). This is because key enzymes and molecules involved in cholesterol homeostasis are closely linked to bone formation. Excess cholesterol may lead to osteoporosis, and cholesterol and its metabolites affect bone homeostasis by regulating the proliferation and stimulation of osteoblasts and osteoclasts (Bao et al., 2023).

Many researchers have investigated the effective mechanisms of statin action on bone (Chamani et al. 2021). The current literature suggests that the effects of statins on bone may involve multiple mechanisms, including proliferation, differentiation, protection of osteoblasts, and reduction of osteoclastogenesis (Oryan et al. 2015). Therefore, statins may play an important role in the clinical management of osteoporosis.

Several studies have shown that disorders of lipid metabolism may be one of the pathogenic mechanisms of osteoarthritis (Haj-Mirzaian et al. 2019; Kadam et al. 2013). For example, high lipid content in synovial fluid is associated with arthritic changes. In addition, elevated leptin levels in obese individuals are associated with articular cartilage destruction (Abramoff and Caldera 2020). Therefore, lipid-lowering drugs may play a role in altering the progression and symptoms of osteoarthritis. Existing studies suggest that statins are effective in reducing inflammation in patients with OA (Dancevic and McCulloch 2014; Siddiq et al. 2023). However, there are no randomized controlled trials (RCTs) validating the clinical efficacy of osteoporosis and osteoarthritis with statin therapy. Implementing a traditional RCT is challenging due to practical, cost, and ethical constraints. Although observational studies are more feasible, their confidence in inferring causality from clinical phenotypes is reduced by confounding factors. In addition, cross-sectional data do not address the issue of temporal order between exposure and outcome, which can lead to reverse causality (Gala and Tomlinson 2020).

Mendelian randomization (MR) is a technique that uses genetic variation as the object of study, with instrumental variables (IVs) being genetic variants that are strongly correlated with an exposure factor (e.g., environmental factor,

lifestyle, or medication, etc.) but are not associated with any confounders of the exposure–outcome association, and there is no pathway through which the instrumental variable can influence other outcomes typically refers to single-nucleotide polymorphisms (SNPs), which serve to assess whether the observed relationship between exposure factors and outcomes is consistent with a causal effect (Evans and Davey Smith 2015). MR investigations differ from standard observational clinical research. MR employs single nucleotide polymorphisms SNPs related with exposure events as instrumental variables. This technique adheres to Mendel's law of random assignment, which simulates a "natural" RCT by effectively eliminating external environmental and cultural impacts on behavior or other factors, as well as stabilizing exposure over time. That is, random assignment performed before birth aligns with the causality timeline (Gupta et al. 2017). MR methods effectively address potential confounders and reverse causality (Li et al. 2024). MR analyses can accurately predict the outcome of RCTs in drug studies, facilitating early and systematic drug target screening (Khasawneh et al. 2022). Currently, two-thirds of FDA-approved drugs are supported by genetic evidence (Ochoa et al. 2022).

Methods and materials

Ethics/consent statement

No ethical approval or informed consent was needed as this study was based on previously published articles and public databases.

Research design

This study examined the causal effect of statins on susceptibility to osteoporosis and osteoarthritis by two-sample MR analysis. Statins were the exposure factor, whereas osteoporosis and osteoarthritis were the outcome variables. To conduct an MR study, three key assumptions must be met: first, the selected SNPs must be significantly correlated with statin exposure; second, the SNPs should be independent of potential confounders; and third, the SNPs must be associated with the risk of developing osteoporosis or osteoarthritis only through the mechanism of statin use (Burgess and Thompson 2017).

Data sources

The FinnGen study, version 10, provides summary statistics from the Genome-Wide Association Study (GWAS) focusing on statins. The study had a large cohort of 342,499 participants, including 233,181 cases of European origin and

109,318 controls of European origin (https://www.finngen.fi/en/access_results). In addition, summary statistics for OP were obtained from the MRC-IEU database, with a sample size of 463,010 people of European ancestry, including 1976 cases of European ancestry and 461,034 controls of European ancestry, and for OA from the United Kingdom BioBank (UKB), with a total sample size of 407,746 people of European ancestry (<https://gwas.mrcieu.ac.uk/>).

Screening of ivs

Single-nucleotide polymorphisms (SNPs) were used as IVs and thresholded at a P value of $<5.0 \times 10^{-8}$. To avoid linkage disequilibrium (LD) bias, the LD of significant SNPs associated with exposure factors had to fulfill the following conditional conditions: $r^2 < 0.001$ and a genetic distance of 10,000 kb. In this study, SNPs significantly associated with exposure factors were extracted from the GWAS dataset of the outcome variables and the resulting IVs were examined to extract the following information Influence Allele, Allele Influence Magnitude (beta), Standard Error and p -value. The F statistics is used to test the strength of each IV and is calculated as follows $F = R^2(N-2)/(1-R^2)$, where R^2 is the proportion of the variance in the exposure factor explained by each IV and N is the sample size of the exposure dataset (Chen et al. 2022). When $F > 10$, there is no weak IV bias (Burgess and Thompson 2011).

Statistical analysis

The summary statistics of the exposure and outcome datasets were aligned so that the effect of SNP on exposure and the effect of SNP on outcome corresponded to the same allele. MR analyses were performed using inverse variance weighting (IVW), weighted median, MR-Egger regression, simple mode, and weighted mode methods to infer causality. Among other things, the inverse variance weighting method is a standard approach in MR analyses for estimating the relationship between a single genetic instrumental variable and a phenotype. The contribution of each instrumental variable (e.g., genetic variance) is weighted during the estimation process, and the weighting is based on the inverse of the estimated variance of the effect of each instrumental variable on the exposure variable. Briefly, the effect of each instrumental variable was assigned a weight that was the inverse of its variance. This is done to enhance the precision of the estimates, especially when the precision of the estimates of the effects of different instrumental variables on the exposure variable is inconsistent. The IVW method is a good choice when all instrumental variables are considered valid, i.e., they affect the outcome variable only by influencing the exposure variable and do not affect the outcome through any other means (Burgess et al. 2015; Hartwig et al. 2016).

MR-Egger regression, on the other hand, is a method that incorporates Wald ratios into a meta-regression for estimating causal effects adjusted for any directional multinomial effects. MR-Egger regression is capable of detecting and correcting for cross-sectional confounders, and its intercepts can provide evidence of direct confounding. It allows for the presence of directional genetic pleiotropy in the included instrumental variables and indicates the presence of genetic pleiotropy when the regression intercept is non-zero and the p value is less than 0.05. MR-Egger regression is a good choice when there is a need to assess the Pleiotropy of instrumental variables or when there are doubts about the invalidity of instrumental variables (Bowden et al. 2015; Bowden et al. 2016). We used IVW as the primary method for MR (Zhang et al. 2022), and when each genetic variant satisfied the IVW assumption, the Wald estimates for each single nucleotide polymorphism were combined by means of meta-analysis to obtain an overall estimate (Zheng et al. 2017). The IVW method gives the most reliable results when there is no horizontal polymorphism in IVW. Compared with the IVW method, the weighted median method improved the accuracy of the results, and complementary analyses using simple and weighted models further improved the accuracy of the results (Xiang et al. 2021).

Sensitivity analysis

Horizontal pleiotropy is a violation of the third IV hypothesis when IVs affect outcomes through pathways that do not involve exposure. Violation of this criterion can severely compromise the accuracy of MR, leading to incorrect quantification of causality, reduced statistical efficacy, and type I errors (Verbanck et al. 2018). We used the MR-Egger test to detect potential pleiotropy, provided that any pleiotropic effect of IVs on the results was independent of exposure. MR-Egger intercept p value > 0.05 indicates that horizontal pleiotropy may not exist (Burgess and Thompson 2017). To assess the robustness of the results, we performed leave-one-out sensitivity analyses, with each iteration involving the systematic exclusion of individual SNPs. We generated forest plots to visually explore potential pleiotropy.

Heterogeneity analysis

Heterogeneity refers to the variability between different IVs observed in MR studies. This variability suggests that each IV may contribute to the phenotype under study to a different extent, or it may only have a causal effect on the feature when interacting with other relevant SNPs. Therefore, the p value of the Cochran Q test was used to assess the heterogeneity among MR values. A p value of less than 0.05 indicates the presence of gender heterogeneity. The Mendelian randomized polytomous residuals and outliers (MR-PRESSO) test was used to

detect and correct for horizontal polytomousness by removing outliers (Lin et al. 2022). MR-PRESSO is a method for assessing horizontal polymorphism in multi-instrument Mendelian randomization using genome-wide summary association statistics. It centers on improving the accuracy of causal inferences by detecting and processing horizontal pleiotropy (i.e., the phenomenon of genetic variation affecting multiple phenotypes), and identifying and correcting outliers (outliers). In statistical analyses, residuals refer to the difference between observed and predicted values. Multi-category residuals may refer to residuals that have many different distributions or characteristics. Outliers, on the other hand, are those values that are significantly different from the majority of observed values, possibly due to measurement error, data entry error, or other unknown factors. The MR-PRESSO method consists of three main components: (1) Detection of Horizontal Multiplicity (MR-PRESSO Integral Test): this test is designed to detect the presence of horizontal multiplicity in the data. The significance of horizontal pleiotropy is assessed by calculating the sum of the squared residuals of the effect of each genetic variant (e.g., SNP) with respect to the inverse variance weighted (IVW) result. The larger the sum of squared residuals, the more significant the horizontal pleiotropy. (2) Correction of horizontal pleiotropy by outlier correction (MR-PRESSO outlier test): After identifying genetic variants that may be subject to horizontal pleiotropy, the MR-PRESSO outlier test further analyses whether these variants are outliers. By comparing the distances (i.e., residuals) of individual genetic variants to the line of fit, variants with larger distances are more likely to be outliers. Once outliers are identified, they can be removed from the analysis to correct for the effects of horizontal pleiotropy. (3) Test for significant distortion in causal estimates before and after outlier removal (MR-PRESSO distortion test): this test is designed to examine whether there is a significant difference in the results of the causal estimates before and after the removal of outliers. By comparing the results before and after correction, it is possible to assess the extent to which outliers affect causal inference and to verify the validity of the correction method. Thus, the MR-PRESSO method provides researchers with a more accurate and reliable analytical tool by detecting and addressing these issues (Verbanck et al. 2018). Statistical analyses were performed in this study using *R* (version 4.3.3) and the *R* package (TwoSampleMR and MR-PRESSO). The test level α was 0.05 ($p < 0.05$) and the difference was statistically significant.

Results

Screening results of SNPs

After strictly adhering to the IVs selection criteria, we performed a complex instrumental variable selection process.

First, we selected SNPs significantly associated with statins therapy ($P < 5 \times 10^{-8}$) in the GWAS dataset. Second, we established the chain disequilibrium parameter ($r^2 = 0.001$, $kb = 10,000$), which ultimately resulted in the selection of 71 different SNPs. Third, we extracted data for the above selected SNPs from the summary statistics of the outcome traits osteoporosis and osteoarthritis, respectively, and also performed allelic matching, a harmonization process where SNPs with incongruent alleles and those that were uncorrectable for the ambiguous palindromic SNPs were removed. And MR-PRESSO was used to identify and remove outliers, finally, we screened 30 SNPs about statins associated with OP as IVs; and also 56 SNPs about statins associated with OA as IVs. And, we calculated the F statistics for each SNP, and they all exceeded the noteworthy threshold of 10. This observation demonstrates the substantial instrumental strength of these SNPs, thus ensuring the robustness of our subsequent MR analyses.

Causal relationship between statins and osteoporosis

Table 1 presents the results of the five research methods for the association between statins and OP. Using the random effects model in the IVW approach, the findings suggest a negative association between statins and OP (OR = 0.974, 95% CI 0.951–0.998, $p = 0.037$). The results were dominated by the IVW method, although still other methods, including MR-Egger, weighted median, weighted mode and simple mode methods, did not support this finding. Figure 1 shows a forest plot visualizing the above five methods for the association of statins with OP risk. Figure 2b Scatterplot, negative correlation between statins and reduced risk of OP development, as shown by the MR regression slope. Figure 2a Forest plot, SNPs have a broadly consistent effect on the results, indicating a stability plot of the results. 2c Leave-one-out plot, with exclusion analysis showing that none of the SNPs had a significant bias on the overall effect

Table 1 Mendelian randomization analysis of the causal relationship between statins and osteoporosis risk. Methods

	SNPs	Beta	SE	OR (95% CI)	P
MR-Egger	30	− 0.001	0.0007	0.998 (0.997,1.000)	0.178
Weighted median	30	− 0.0008	0.0004	0.999 (0.998,1.000)	0.062
IVW	30	− 0.001	0.0003	0.998 (0.998,0.999)	0.0002
Simple mode	30	− 0.0006	0.0008	0.999 (0.997,1.001)	0.478
Weighted mode	30	− 0.0008	0.0005	0.999 (0.998,1.000)	0.120

exposure	outcome	nsnp	method	pval	OR(95% CI)
Statin medication	Osteoarthritis	56	MR Egger	0.058	0.965 (0.932 to 1.000)
	Osteoarthritis	56	Weighted median	0.028	0.963 (0.931 to 0.996)
	Osteoarthritis	56	Inverse variance weighted	0.037	0.974 (0.951 to 0.998)
	Osteoarthritis	56	Simple mode	0.291	0.969 (0.915 to 1.027)
	Osteoarthritis	56	Weighted mode	0.040	0.967 (0.937 to 0.998)
Osteoporosis	Osteoporosis	30	MR Egger	0.178	0.999 (0.997 to 1.000)
	Osteoporosis	30	Weighted median	0.062	0.999 (0.998 to 1.000)
	Osteoporosis	30	Inverse variance weighted	<0.001	0.999 (0.998 to 0.999)
	Osteoporosis	30	Simple mode	0.475	0.999 (0.998 to 1.001)
	Osteoporosis	30	Weighted mode	0.121	0.999 (0.998 to 1.000)

Fig. 1 Forest plots of five outcomes from Mendelian randomization analyses of statins for osteoporosis and osteoarthritis in inverse variance weighted, weighted median, MR-Egger regression, simple mode and weighted mode methods. Effect sizes are shown by black squares, 95% confidence intervals are black horizontal solid lines, and the vertical dashed line in the middle is the null line, i.e., OR = 1; when the black horizontal solid line intersects the null line it indicates that there is no causal relationship between the study exposure factors and

the outcome association. When the horizontal line is to the right of the null line, it indicates a positive relationship between the study factor and the occurrence of the outcome event; when the horizontal line is to the left of the null line, it indicates a negative relationship between the study factor and the occurrence of the outcome event. $p < 0.05$ indicates statistical significance. *CI* confidence interval, *nSNP* number of single nucleotide polymorphism, *OR* odds ratio

of statins on OP. Figure 2d Funnel plot showing symmetry, excluding multiple effects.

Heterogeneity test and sensitivity analysis

Sensitivity analyses were subsequently conducted to assess the robustness of the main results, and there was no heterogeneity in the Heterogeneity test (IVW) analyses (Cochran's $Q = 31.29$, $P = 0.351$) and MR-Egger analyses (Cochran's $Q = 31.26$, $P = 0.308$). In addition, the MR-Egger intercept test ($P = -1.266 \times 10^{-5}$) showed that our analyses did not find significant evidence of horizontal pleiotropy, with all P values exceeding the 0.05 threshold (Table 3).

Causal relationship between statins and osteoarthritis

Table 2 presents the results of the five research methods for the association between statins and OA. Using the random effects model in the IVW approach, a negative association between statins and OA was observed (OR = 0.998, 95% CI 0.998–0.999, $P = 0.0002$); and the weighted median (OR = 0.963, 95% CI 0.931–0.996, $P = 0.028$) and weighted modal approaches (OR = 0.967, 95% CI 0.937–0.998, $P = 0.039$), consistently supported this finding. Figure 1 shows a forest plot visualizing the above five methods for the association of statins with OA risk. In addition, the forest plot, leave-one-out analysis plot, and funnel plot showed that the effect of each SNP on the results was generally consistent, indicating the stability of our results (Fig. 3).

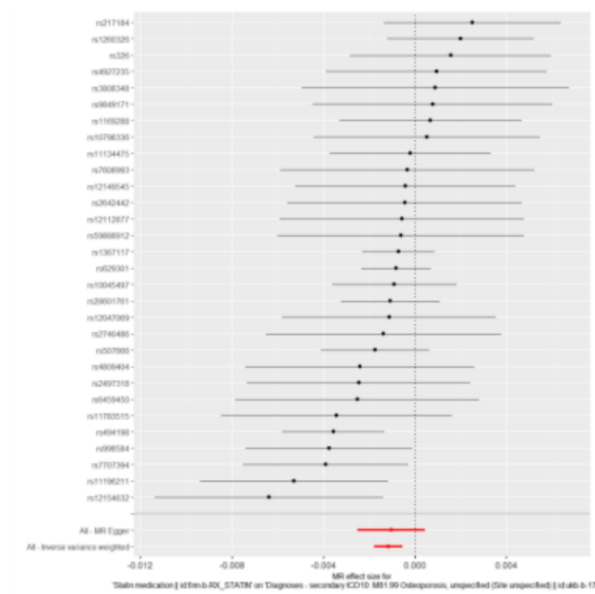
Heterogeneity test and sensitivity analysis

A combined analysis of heterogeneity, including the Cochran's Q test for heterogeneity and the MR-Egger regression for multi-directionality (Table 3) showed that there was no heterogeneity in the IVW analyses (Cochran's $Q = 72.07$, $P = 0.051$) and MR-Egger analyses (Cochran's $Q = 72.73$, $P = 0.055$) (Table 3). In addition, the horizontal pleiotropy test (MR-Egger) intercept test (Intercept = 0.1117) showed that no significant evidence of horizontal pleiotropy was found, with P values exceeding the 0.05 threshold (Table 3). The results of all these tests indicate the robustness of the MR findings.

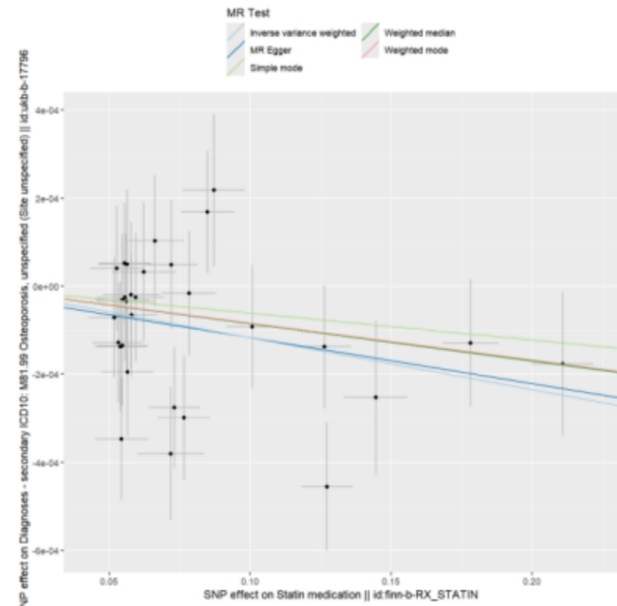
Discussion

Exploring the causal relationship between exposure factors and osteoporosis and osteoarthritis disease is essential for disease prevention and the development of improved treatment strategies. In this two-sample Mendelian randomization study, we revealed that statins reduce the risk of osteoporosis and osteoarthritis in a European population, with a negative correlation between the two from a genetic prediction perspective.

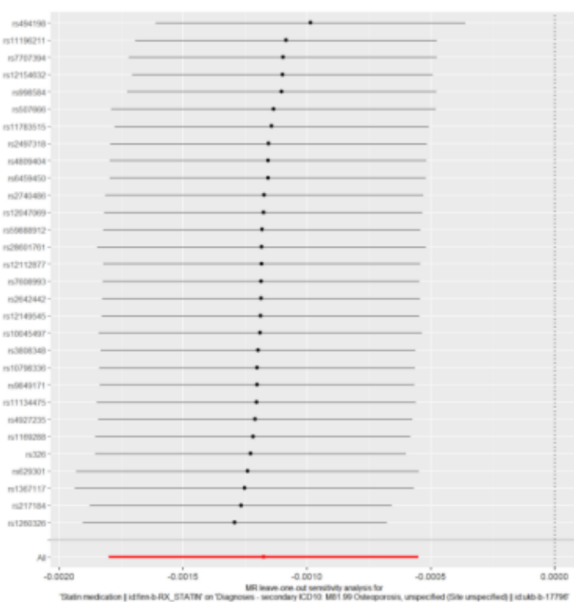
Previous studies have highlighted the positive shadow of statins on cartilage synthesis and metabolism (Bao et al. 2023; Chamani et al. 2021; Ren and Zhou 2024). Our study further demonstrates the positive effects of statins in preventing OA and OP. Several previous observational studies have explored the relationship between statins and osteoporosis and osteoarthritis. In an experimental animal study on



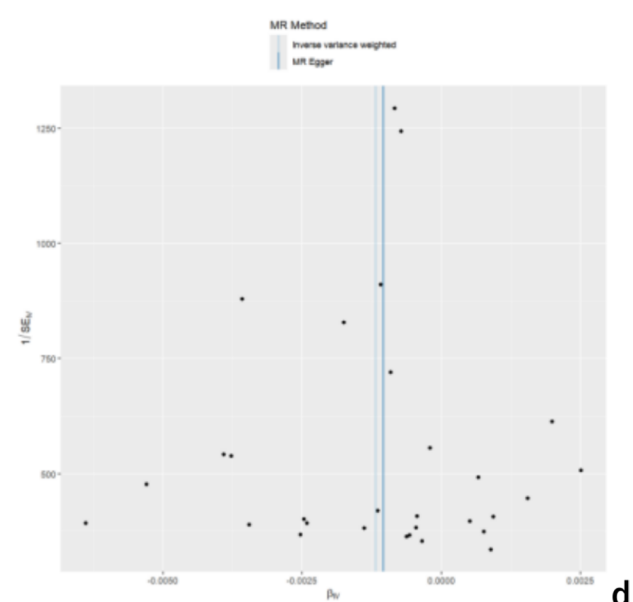
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osteoarthritis, it was found that intra-articular injection of statins reduced arthritis in a rabbit model of osteoarthritis (Goto et al. 2017). In addition, statins prevented cartilage degeneration after IL-1 β stimulation and induced anti-inflammatory effects by stimulating the STAT1-caspase-3 signaling pathway (Saberianpour et al. 2022). In addition, statins also attenuate the inflammatory response induced by sodium iodoacetate in a rat joint model (Pathak et al. 2015). In terms of clinical management of osteoporosis and osteoarthritis, previous studies have found that long-term use of statins reduces the incidence of osteoarthritis and slows its

progression (Mohajer et al. 2024). Furthermore, an observational study conducted by Dutch scholars found that long-term use of statins reduced the incidence of osteoarthritis and slowed its progression (Clockaerts et al. 2012). It has also been demonstrated that statins do not reduce the risk of hospitalization for osteoarthritis in the hip and knee. In addition, a Korean study demonstrated that statin use was associated with a significantly lower risk of osteoporotic fracture in an elderly population (Seo et al. 2023). In a population-based cohort study in Taiwan, researchers found that statin use was associated with a reduced risk of osteoporosis in

Fig. 2 a Forest plot, Each of its horizontal solid lines responds to a single SNP, the results estimated using the Wald ratio method, the solid line is entirely to the left of 0, indicating that the result estimated from this SNP, is that statins reduce the risk of developing OP, the solid line is entirely to the right of 0, indicating that the result estimated from this SNP, is that statins increase the risk of developing OP, and those results that cross over 0 indicate that it is not significant, but the results for a single SNP are not robust because statins here are influenced by multiple SNPs, so the results need to be taken together, which is the red line at the bottom, and the red line below that shows that taking statins reduces the risk of developing OP. **b** Scatter plot, representing the results of MR analysis between exposure factors and outcome factors, with the x-axis representing the effect of SNP on statins and the y-axis representing the effect of SNP on OP. The slope is less than 0, which means that the exposure factor is a favorable factor for the outcome, and the different colored lines represent the different algorithms, and the results show that the lines for the different algorithms are overall slanted downwards, which means that statins can reduce the risk of developing OP. **c** Sensitivity analysis, Each entry represents the result of MR analysis excluding a subset of SNPs for that SNP, and the corresponding horizontal coordinate is its effect size estimate (i.e., the effect size estimate after the leave-one-out method). As shown in Fig. 2c above, the overall error line did not change much after excluding each SNP, suggesting that the results of statins to reduce the incidence of OP are reliable. **d** Funnel plot, where each point represents the effect size of an SNP, the horizontal coordinate usually indicates the effect size, and the vertical coordinate indicates the precision (standard error), the observation that the points on the left and right sides of the IVW line are roughly symmetrical suggests that there is no heterogeneity of SNPs, and that the results of statins to reduce the risk of OP morbidity are reliable

both men and women (Lin et al. 2018a, b). Whether statins can treat osteoporotic osteoarthritis remains controversial in the existing phase of research (Lin et al. 2018a, b; Wang et al. 2021). However, the present MR study provides a theoretical basis for previous observational studies, and the detection of specific related genes can predict their treatment effects. It also provides important new evidence for the rational use of statins in patients with osteoporosis and osteoarthritis, which is important for disease prevention and the development of improved treatment strategies. However, some adverse effects of statins, such as gastrointestinal reactions, cognitive dysfunction, hepatic dysfunction, muscle weakness and pain, and in severe cases, rhabdomyolysis; although the incidence of the above symptoms is still low, these adverse effects may limit its use in some special populations (Attardo et al. 2022; Bellosta and Corsini 2018). Randomized controlled trials (RCTs) are needed to validate the efficacy of statins in the treatment of clinical OA and to avoid the serious adverse effects associated with oral statins.

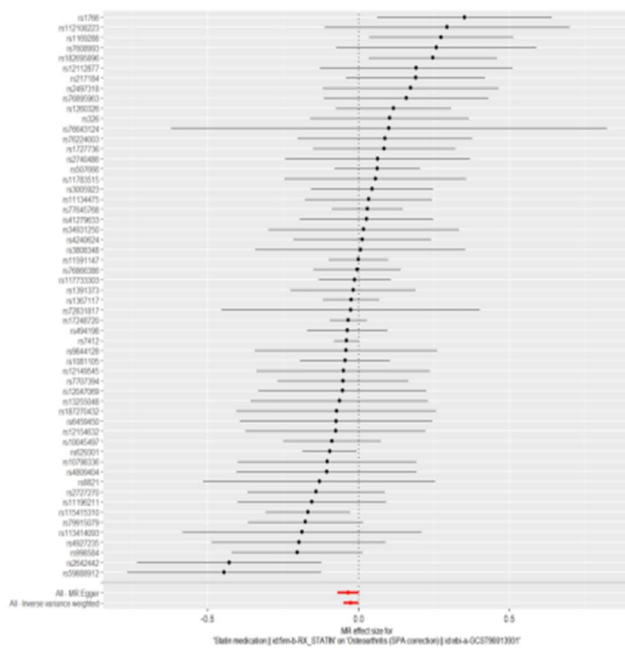
For the application of statins to reduce the risk of osteoporosis and osteoarthritis, we can explain this causal relationship on the basis of existing biological mechanisms. The main pathogenesis of osteoporosis is a disorder of bone metabolism characterized by an imbalance between bone formation and resorption, which ultimately leads to a significant decrease in bone mass and density (Zou et al. 2020).

Osteoarthritis is characterized by an imbalance between altered homeostasis (anabolic) and catabolic (degenerative) metabolism within the articular cartilage mediated by the innate immune system, leading to synovial membrane hyperplasia, degeneration of the articular cartilage, disruption of the structural and functional integrity of the cartilage's extracellular matrix, subchondral sclerosis, and formation of osteoclasts (Mukherjee and Das 2024). The mechanisms by which statins modulate osteoporosis and osteoarthritis have not been fully elucidated. However, several possible mechanisms have been proposed to elucidate the pathways by which statins affect bone metabolism and inflammation. First, statins increase bone morphogenetic protein-2 (BMP-2) through the Ras-PI3K-Akt/MAPK signaling pathway (Ruan et al. 2012), and BMP-2 induces osteoblast differentiation through Runt-related transcription factor 2 (Runx2) (Jensen et al. 2010; Nguyen et al. 2022). Second, statins inhibit the mevalonate pathway, preventing the synthesis of downstream products. As a result, FPP and GGPP synthesis is blocked (Goldstein and Brown 1990), which negatively regulates osteoblast differentiation (Lee et al. 2022; Weivoda and Hohl 2011). Third, statins inhibit osteoblast apoptosis via the TGF β /Smad3 pathway (Heldin, Miyazono, and ten Dijke 1997). Fourth, statins inhibit osteoclastogenesis via the OPG/RANKL/RANK pathway (Tsubaki et al. 2012). Based on the above mechanisms of action, statins are thought to be associated with bone anabolism. We also confirmed a causal relationship between statins use and reduced risk of osteoporosis and osteoarthritis in terms of genetic correlation and Mendelian randomization analysis. These results enhance our understanding of the role of statins in osteoporosis and osteoarthritis prevention and suggest that statins may be an effective prevention strategy for individuals at risk of osteoporosis and osteoarthritis.

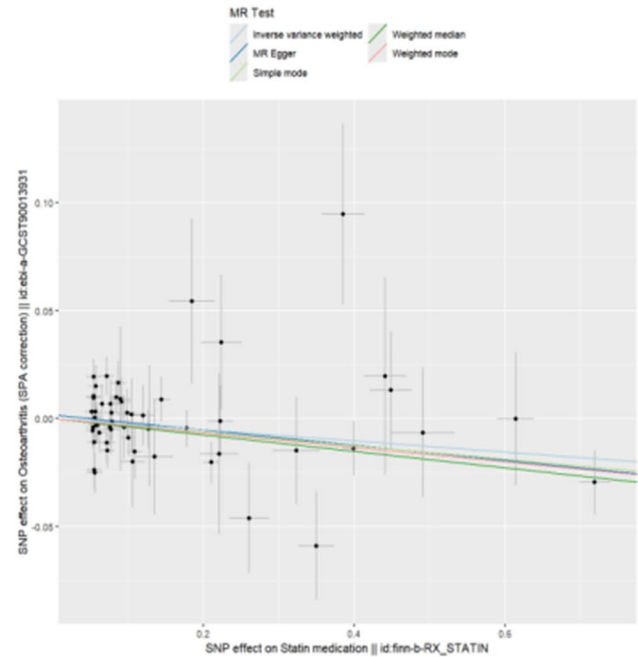
Our analyses have several advantages. First, the MR analysis relies on the random assignment of genetic variation during conception and is therefore less susceptible to individual selection bias or behavioral interference. This simulation condition, similar to that of a randomized clinical

Table 2 A Mendelian randomization analysis of the causal relationship between statins and risk of osteoarthritis

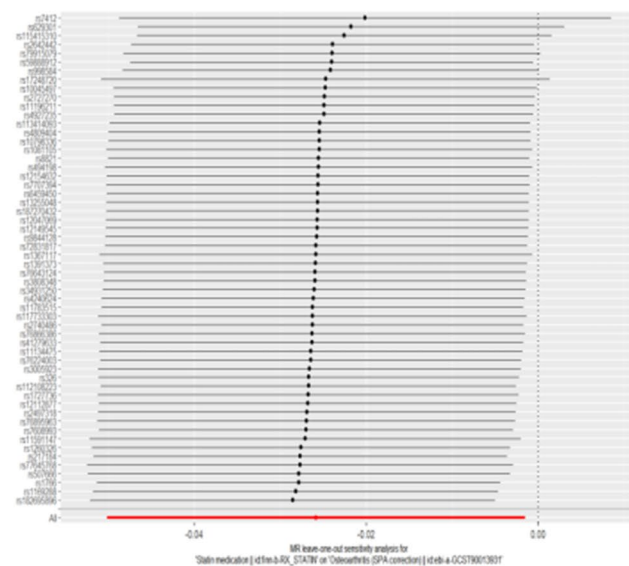
Methods	SNPs	Beta	SE	OR (95% CI)	P
MR-Egger	56	- 0.035	0.018	0.965 (0.932,1.000)	0.058
Weighted median	56	- 0.038	0.017	0.963 (0.931,0.996)	0.028
IVW	56	- 0.026	0.012	0.974 (0.951,0.998)	0.037
Simple mode	56	- 0.031	0.029	0.969 (0.915,1.027)	0.291
Weighted mode	56	- 0.035	0.015	0.967 (0.937,0.998)	0.039



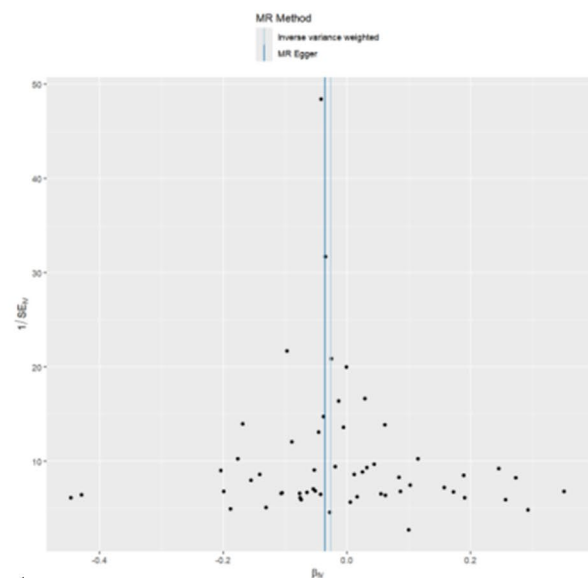
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trial, ensures more reliable results for causal inference. Second, the use of exposure and outcome datasets from different sources minimizes the bias inherent in weak instrumental variable methods, thereby increasing the validity of two-sample MR analyses. Third, strict criteria for the selection of instrumental variables were developed, allowing only SNPs significantly associated with statins therapy and meeting

the three core assumptions of MR analysis to be used as instruments. In addition, genetic variants were dispersed across different chromosomes, thus mitigating the potential impact of gene–gene interactions on effect estimates. Fourth, to assess the validity of the instrumental variable hypotheses, we used a range of different analytical methods, including heterogeneity tests, cross-sectional multivariate

Fig. 3 a Forest plot, Each horizontal solid line corresponds to an SNP. The results estimated using the Wald ratio method have a solid line exactly to the left of 0 indicating that statins reduce the risk of OA according to that SNP, a solid line exactly to the right of 0 indicating that statins increase the risk of OP, and those results crossing 0 indicating that they are not significant. increases the risk of developing OA, and those crossing 0 indicate that it is not significant, but the results for a single SNP are not reliable because statins are affected by more than one SNP, so these results need to be considered together, which is the red line at the bottom, and the red line below indicates that taking a statin reduces the risk of developing OA. **b** Scatter plot, Represents the results of MR analysis between exposure factors and outcome factors, the x-axis represents the effect of SNP on statins and the y-axis represents the effect of SNP on OA. The slope is less than 0, which means that the exposure factor is a favorable factor for the outcome, and the different colored lines represent the different algorithms, and the results show that the lines for the different algorithms are overall slanted downwards, which means that statins can reduce the risk of OA. **c** Sensitivity analysis, Each entry represents the result of MR analysis excluding a subset of SNPs for that SNP, and the corresponding horizontal coordinate is its effect size estimate (i.e., the effect size estimate after the leave-one-out method). As shown in (c) above, the overall error line did not change much after excluding each SNP, suggesting that the results of statins to reduce the incidence of OA are reliable. **d** Funnel plot, where each point represents the effect size of an SNP, the horizontal coordinate usually indicates the effect size, and the vertical coordinate indicates the precision (standard error), the observation that the points on the left and right sides of the IVW line are roughly symmetrical suggests that there is no heterogeneity of SNPs, and that the results of statins to reduce the risk of OA morbidity are reliable

assessment and skimming analyses. Finally, in addition to Mendelian randomization analyses, we fully elaborate on possible underlying mechanisms. This further enhances the validity of the results of our Mendelian randomization analyses compared to previous similar studies (Yu et al. 2024).

There are several limitations to this study. First, our focus on excluding SNPs associated with known confounders may have overlooked other unidentified confounders that may influence the relationship between statins and OA. This limitation highlights the need for further research to identify these potential confounders. Second, although statins have shown potential benefit in the prevention of osteoarthritis, specific treatment regimens and dosages need to be determined by further research and clinical practice. Third, our study population was of exclusively European origin. This factor limits the generalizability of our findings to other populations due to different disease patterns in different ancestral backgrounds. Fourth, there are several statins, each with different pharmacokinetic and pharmacodynamic properties. These differences, such as variations in absorption rates and tissue binding affinities, may affect the therapeutic efficacy of statins. Therefore, our future research program includes a focus on individual statins or different statins classes to understand their effects in more detail.

Conclusions

This study, at the genetic level, demonstrates that statins can reduce the risk of osteoporosis and osteoarthritis in the population in a causal manner. Therefore, this study provides potential evidence for the future use of statins in anti-osteoporosis and osteoarthritis treatment. However, further research is essential to confirm our findings. This includes the conduct of large-scale RCTs focused on explicitly assessing the beneficial effects and clear mechanisms of action of statins in the treatment of osteoporosis and osteoarthritis.

Table 3 Heterogeneity test and horizontal pleiotropy test

Exposure	Outcome	Heterogeneity test (MR-Egger)			Heterogeneity test (IVW)			Horizontal pleiotropy test (MR-Egger)	
		Cochran's Q	Q_df	P	Cochran's Q	Q_df	P	Intercept	P
Statins	OP	31.26	28	0.308	31.29	29	0.351	- 1.266E- 05	0.848
Statins	OA	72.07	54	0.051	72.73	55	0.055	0.1117	0.483

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Declarations

Conflict of interest The authors declare that they have no competing interests.

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